

Assignment 9: Time Series Analysis

Aadya Shukla

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/872L/EDE_Spring2026
```

```
library(zoo)
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##    as.Date, as.Date.numeric
```

```
library(trend)  
library(Kendall)  
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':  
##    method      from  
##    as.zoo.data.frame zoo
```

```
here()
```

```
## [1] "/home/guest/872L/EDE_Spring2026"
```

```
mytheme <- theme_light(base_size = 14) +  
  theme(axis.text = element_text(color = "black", face = "bold"),  
        axis.title = element_text(face = "bold"),  
        legend.position = "right")  
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1  
Ozone2010 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),  
  stringsAsFactors = TRUE)  
Ozone2011 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),  
  stringsAsFactors = TRUE)  
Ozone2012 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),  
  stringsAsFactors = TRUE)  
Ozone2013 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),  
  stringsAsFactors = TRUE)  
Ozone2014 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),  
  stringsAsFactors = TRUE)  
Ozone2015 <- read.csv(  
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
```

```

  stringsAsFactors = TRUE)
Ozone2016 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
  stringsAsFactors = TRUE)
Ozone2017 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
  stringsAsFactors = TRUE)
Ozone2018 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
  stringsAsFactors = TRUE)
Ozone2019 <- read.csv(
  here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
  stringsAsFactors = TRUE)

GaringerOzone <- rbind(
  Ozone2010, Ozone2011, Ozone2012, Ozone2013, Ozone2014,
  Ozone2015, Ozone2016, Ozone2017, Ozone2018, Ozone2019)

```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```

# 3
GaringerOzone$Date <- mdy(GaringerOzone$Date)

# 4
GaringerOzone_edited <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(as.Date("2010-01-01"), as.Date("2019-12-31"), 1))
colnames(Days) <- "Date"

# 6
GaringerOzone <- left_join(Days, GaringerOzone_edited)

## Joining with 'by = join_by(Date)'

```

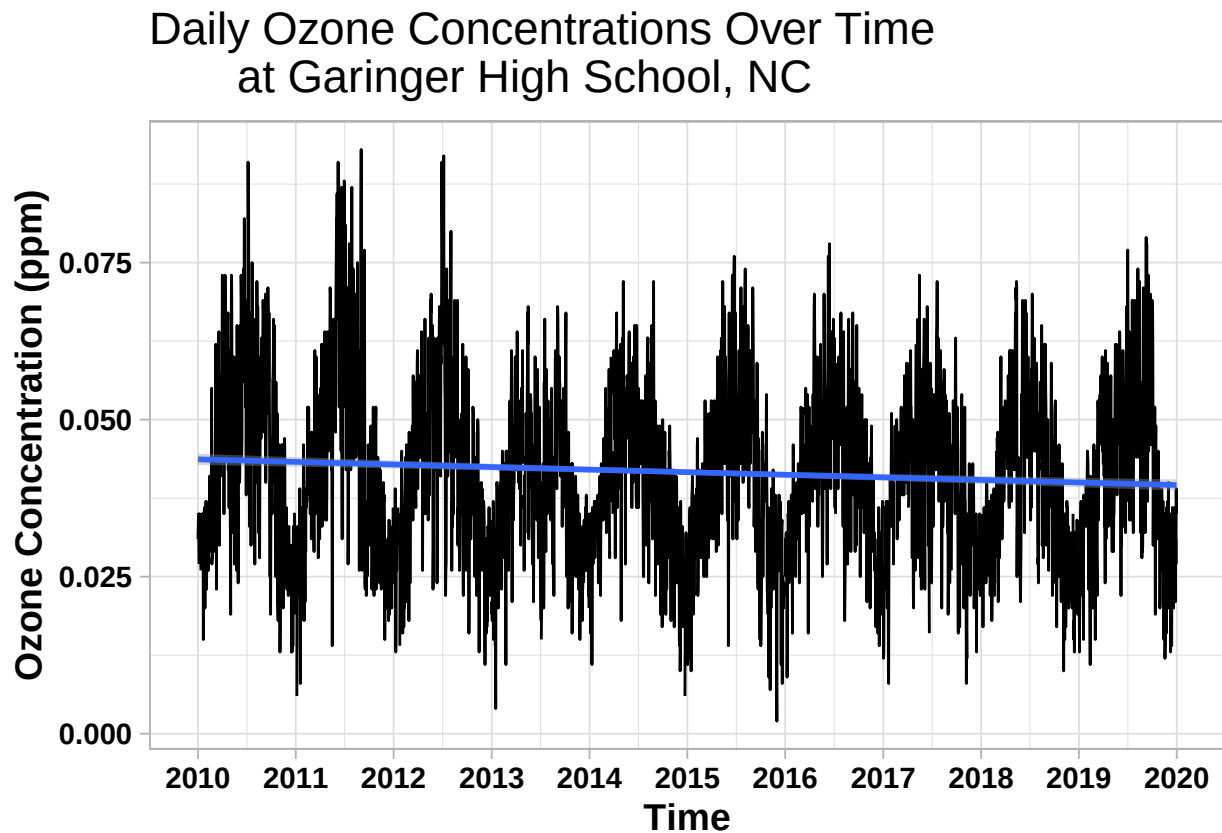
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(GaringerOzone,
  aes(y = Daily.Max.8.hour.Ozone.Concentration, x = Date)) +
  geom_line() +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(y = "Ozone Concentration (ppm)", x = "Time",
    title = "Daily Ozone Concentrations Over Time
    at Garinger High School, NC")

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: Yes, based on this graph, there seems to be a trend of ozone concentration increasing and decreasing cyclically, with an overall linear trend where ozone concentrations seem to be decreasing very slightly.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone_clean <-
  GaringerOzone %>%
  mutate(Ozone.Concentration.clean = zoo::na.approx(
    Daily.Max.8.hour.Ozone.Concentration) )
```

Answer: Using a linear interpolation was best given that the data had very few NA values spread across intermittently and the available ozone values were following a simple steady progression of values. To keep the data within the range of the available values and to fill in minor gaps, the linear method was chosen.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone_clean %>%
  mutate(Month = month(Date), Year = year(Date)) %>%
  mutate(First.Month = make_date(Year, Month, 1)) %>%
  group_by(First.Month) %>%
  summarise(Mean.Ozone = mean(Ozone.Concentration.clean))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

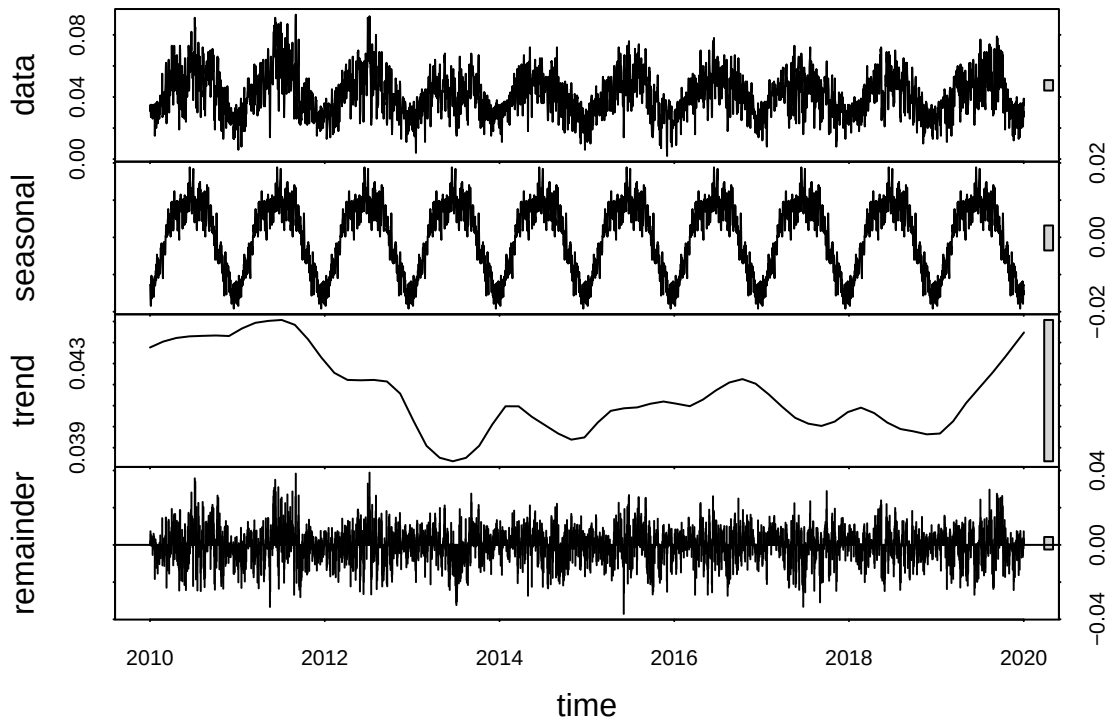
```
#10
GaringerOzone.daily.ts <- ts(
  GaringerOzone_clean$Ozone.Concentration.clean,
  start = c(2010,1), frequency = 365)

GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$Mean.Ozone,
  start = c(2010,1), end = c(2019,12), frequency = 12)
```

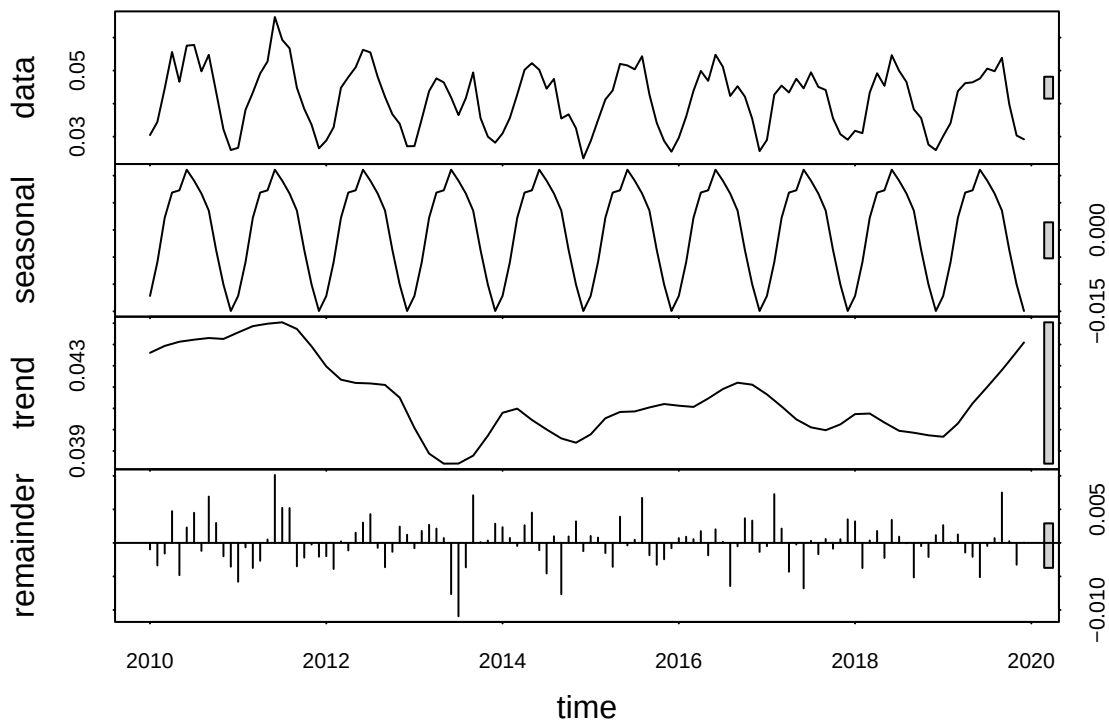
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

#11

```
GaringerOzone.daily_decomposed <- stl(  
  GaringerOzone.daily.ts, s.window = "periodic")  
  
plot(GaringerOzone.daily_decomposed)
```



```
GaringerOzone.monthly_decomposed <- stl(  
  GaringerOzone.monthly.ts, s.window = "periodic")  
  
plot(GaringerOzone.monthly_decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12

```
GaringerOzone.monthly_trend <- Kendall::SeasonalMannKendall(
  GaringerOzone.monthly.ts)
```

```
GaringerOzone.monthly_trend
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
summary(GaringerOzone.monthly_trend)
```

```
## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

```
GaringerOzone.monthly_trend2 <- trend::smk.test(GaringerOzone.monthly.ts)
```

```
GaringerOzone.monthly_trend2
```

```
##
```

```
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -77 1499
```

```
summary(GaringerOzone.monthly_trend2)
```

```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS      tau      z Pr(>|z|)
## Season 1:  S = 0   15  125  0.333  1.252  0.21050
## Season 2:  S = 0   -1  125 -0.022  0.000  1.00000
## Season 3:  S = 0   -4  124 -0.090 -0.269  0.78762
## Season 4:  S = 0  -17  125 -0.378 -1.431  0.15241
## Season 5:  S = 0 -15  125 -0.333 -1.252  0.21050
## Season 6:  S = 0 -17  125 -0.378 -1.431  0.15241
## Season 7:  S = 0 -11  125 -0.244 -0.894  0.37109
## Season 8:  S = 0   -7  125 -0.156 -0.537  0.59151
## Season 9:  S = 0   -5  125 -0.111 -0.358  0.72051
## Season 10: S = 0 -13  125 -0.289 -1.073  0.28313
## Season 11: S = 0 -13  125 -0.289 -1.073  0.28313
## Season 12: S = 0  11  125  0.244  0.894  0.37109
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The seasonal Mann-Kendall test is the only analysis that accounts for seasonality. Our monthly ozone series data shows a rhythmic seasonal variation in the values, thereby requiring a seasonal test like the seasonal Mann-Kendall.

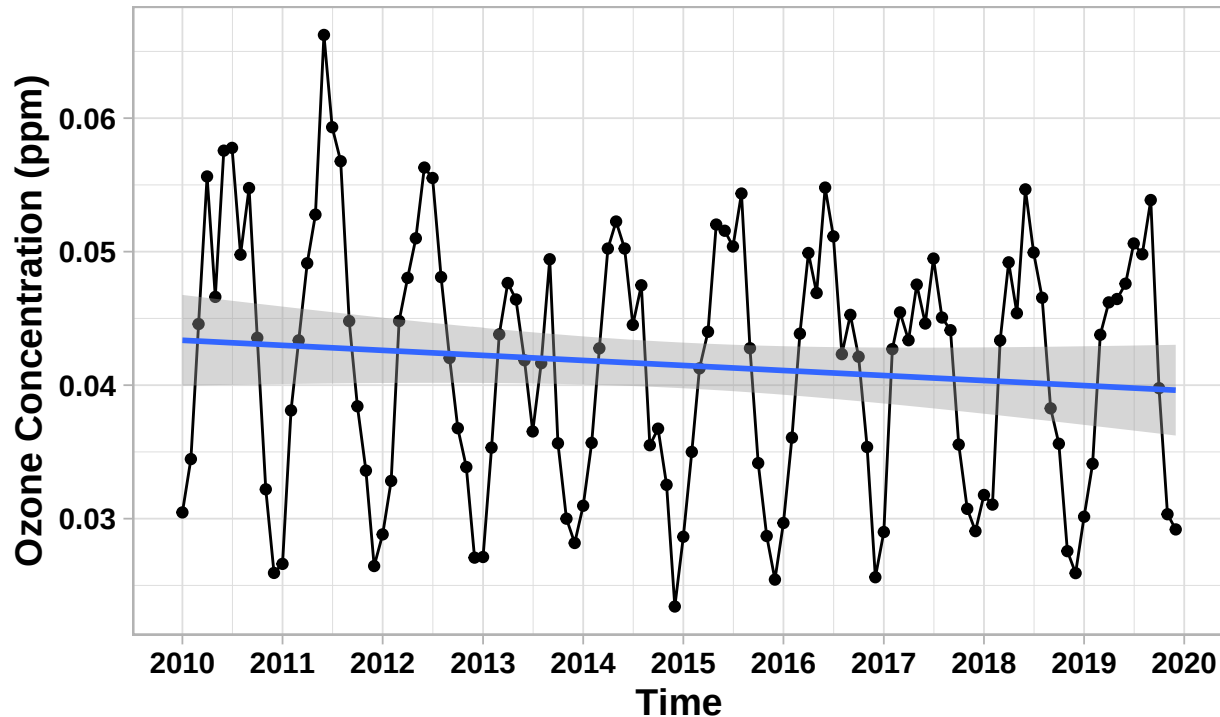
13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(y = Mean.Ozone, x = First.Month)) +
  geom_point() +
  geom_line() +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  geom_smooth(method = lm) +
  labs(x = "Time", y = "Ozone Concentration (ppm)",
       title = "Mean Monthly Ozone Concentrations over Time
at Garinger High School, NC")
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

Mean Monthly Ozone Concentrations over Time at Garinger High School, NC



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The Mann-Kendall seasonal analysis resulted in a p-value which is slightly lower than the 0.05 threshold ($p = 0.046724$). This significance, however, is extremely borderline. Thus, there is an overall monthly monotonic trend in the ozone concentrations over time. This trend seems to be an overall negative trend ($S = -77$). Looking at the individual seasons/months, there doesn't seem to be a high variance as values range from -17 to 15 and the p-values for none of the seasons are significant ($p > 0.05$), indicating that we cannot reject the null hypothesis that there is no monotonic trend in the data for the individual seasons but we can reject the hypothesis for the range of 2010-2019 ($p < 0.05$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

#15

```
GaringerOzone.nonseasonal <- GaringerOzone.monthly.ts - GaringerOzone.monthly_decomposed$time.series[,1]
GaringerOzone.nonseasonal
```

##	Jan	Feb	Mar	Apr	May	Jun
## 2010	0.04263190	0.04041003	0.04234881	0.04875492	0.03932081	0.04647348
## 2011	0.03877706	0.04405289	0.04112300	0.04225492	0.04548211	0.05514015
## 2012	0.04098674	0.03877333	0.04257462	0.04115492	0.04370791	0.04520681
## 2013	0.03929319	0.04126717	0.04157462	0.04077159	0.03912727	0.03077348
## 2014	0.04313190	0.04162432	0.04052623	0.04335492	0.04496598	0.03914015
## 2015	0.04080932	0.04094575	0.03902623	0.03712159	0.04474017	0.04047348
## 2016	0.04184158	0.04201471	0.04162300	0.04302159	0.03961114	0.04370681
## 2017	0.04116416	0.04864217	0.04321978	0.03648826	0.04024017	0.03352348
## 2018	0.04393835	0.03699932	0.04112300	0.04232159	0.03809501	0.04357348
## 2019	0.04230932	0.04005289	0.04154236	0.03932159	0.03915952	0.03650681
##	Jul	Aug	Sep	Oct	Nov	Dec
## 2010	0.04871023	0.04307797	0.05120815	0.04725211	0.04226527	0.04087082
## 2011	0.05025862	0.05007797	0.04124148	0.04212308	0.04366527	0.04138695
## 2012	0.04645216	0.04140056	0.03847481	0.04047792	0.04393193	0.04201598
## 2013	0.02746829	0.03494894	0.04587481	0.03934888	0.04006527	0.04311275
## 2014	0.03545216	0.04078765	0.03194148	0.04044566	0.04259860	0.03835469
## 2015	0.04132313	0.04765862	0.03920815	0.03786501	0.03876527	0.04037082
## 2016	0.04208120	0.03562636	0.04170815	0.04583275	0.04543193	0.04054824
## 2017	0.04041991	0.03836830	0.04055815	0.03925211	0.04079860	0.04399985
## 2018	0.04087152	0.03985217	0.03470815	0.03931662	0.03763193	0.04085469
## 2019	0.04154894	0.04311023	0.05030815	0.04347792	0.04039860	0.04412888

#16

```

GaringerOzone.nonseasonal.trend <- Kendall::MannKendall(
  GaringerOzone.nonseasonal)

GaringerOzone.nonseasonal.trend

```

```
## tau = -0.165, 2-sided pvalue =0.0075402
```

```
summary(GaringerOzone.nonseasonal.trend)
```

```

## Score = -1179 , Var(Score) = 194365.7
## denominator = 7139.5
## tau = -0.165, 2-sided pvalue =0.0075402

```

Answer: The non-seasonal Mann-Kendall test supports our previous seasonal results. For the overall trend, the p-value reduced by a lot ($p = 0.0075402$). This indicates that the data does have an overall significant trend that is decreasing over time. However, when seasonality comes into play, the individual seasonal trends are barely or not at all present.